

Full Name: SOLUTIONS

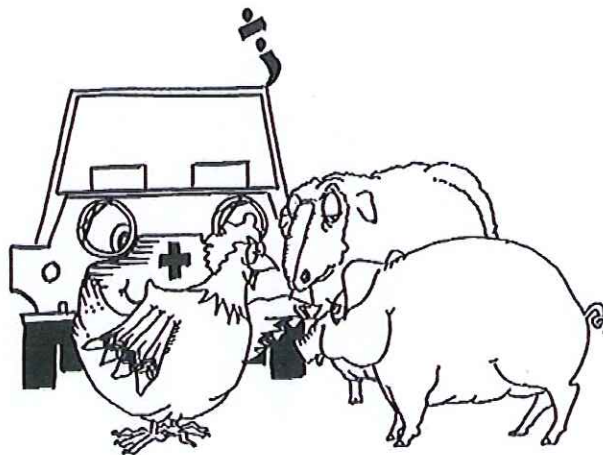


Ellenbrook
CHRISTIAN COLLEGE

**MATHEMATICS
YEAR 8**

Problem 4 – The Farmer's Puzzle

Semester 2 2014



Information for Students

Working time:	30 minutes
Marks available:	12 marks

How to Solve a Problem

READ & UNDERSTAND

Do I understand what the problem is asking?
Discuss.

PLAN A STRATEGY

Select a strategy, or strategies, from the Strategy board. Plan how you intend solving the problem.

DO IT

Try out your idea.

CHECK & REFLECT

Did it work out? If so, reflect on the activity.
Perhaps publish your work for others to see.
If not, go back to step one.

STRATEGY BOARD**HAVE I SEEN A SIMILAR PROBLEM?****GUESS & CHECK****LOOK FOR A PATTERN****ACT IT OUT****DRAW A PICTURE OR GRAPH****MAKE A MODEL****MAKE A LIST OR TABLE****TRY ALL POSSIBILITIES****WORK BACKWARDS****SOLVE A SIMPLER
RELATED PROBLEM****USE A SUM OR
NUMBER SENTENCE****SEEK AN EXCEPTION****BREAK IT INTO MORE
MANAGEABLE PARTS**

The Farmer's Puzzle

A hundred dollars a farmer paid
For a hundred animals 'through the trade'.
Sheep were ten dollars, pigs were two,
Hens were fifty cents. What did she do?

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Task 1 (4 marks)

1. How many hens can the farmer buy for a \$100?
Explain why he could not have bought just hens.

[2]

hens 50¢ each

$$100 \div 0.5 = 200$$

∴ 200 hens can be purchased ✓

can only buy 100 animals
∴ The farmer cannot buy only hens. ✓

2. What would she pay for a 100 hens?

[1]

$$100 \times 0.5 = 50$$

∴ \$50 for 100 hens ✓

3. Explain why she cannot buy just pigs or sheep.

[1]

will not buy enough animals.

only pigs $\$100 \div \$2 = 50$ pigs (not enough)

only sheep $\$100 \div \$10 = 10$ sheep (not enough)

✓ acceptable response.

Task 2 (6 marks)

One of the strategies you can choose is to make a table and do a guess and check.

1. In the table below, why do we start with 96 hens and not 99? [1]

$99 \times 0.5 = \$49.50$ leaving $\$50.50$ for other animals.
Cannot spend the $\$50.50$... not possible

2. a. Use the table below or your own strategy (on next page) to solve "The Farmer's puzzle". [4]

Hens \$0.50	Money Remaining	Sheep \$10	Pigs \$2	Animals 100	Money \$100	Problem Solved Yes/No
96	\$52	1	21	Too many	Yes	No
94	53	5	1	✓	No, \$99	No
92	54	5	2	Too few	Yes	No
90	55				No	
88	56	5	3	Too few	Yes	No
* 88	56	4	8	✓	Yes	Yes
!	↑ only expect one solution here for marks.					
!	↓					
* 72	64	1	27	✓	Yes	Yes

✓ 4 mks good attempt; even if no solution shown.

(1 mk, 1 line attempt; 2 mks, multiple lines but incorrect; 3 mks, good but not great)

2. b. My solution

[1]

88 hens, 4 sheep, 8 pigs.
or 72 1 27



Space for other strategies, if needed.

Extension

So, there is at least one solution.

Are there any more? Demonstrate or Explain.

[2]

You may use the back of this page if more space is required.

- Continue table
- Spreadsheet

2 marks for the second solution

- No marks if no other solution
- 2nd Solⁿ mentioned previously is still 2 marks

End of Problem

1 mark if only ideas